Subspaces and Rank

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Rank

Definition 0.1 (Linearly independent vectors)**.** Vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ of the same length p are **linearly independent** [1] if the only numbers $c_1, \ldots, c_k$ with

$$c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k = \underset{\sim}{0}$$

are

$$\begin{aligned} c_1 &= \cdots \\ &= c_k \\ &= 0. \end{aligned}$$

Vectors that are not linearly independent are **linearly dependent**.

Example 0.1 (Independent and dependent pairs of vectors)**.**

- $\underset{\sim}{v}_1 = (1, 0)$ and $\underset{\sim}{v}_2 = (1, 1)$ are linearly independent: $c_1 \underset{\sim}{v}_1 + c_2 \underset{\sim}{v}_2 = (c_1 + c_2, c_2)$, which is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$.
- $\underset{\sim}{v}_1 = (1, 2)$ and $\underset{\sim}{v}_2 = (2, 4)$ are not: $2\underset{\sim}{v}_1 - \underset{\sim}{v}_2 = \underset{\sim}{0}$.

Exercise 0.1 (Testing whether two features and their total are linearly independent)**.** A table of data that stores two features and also their total holds redundant information, which the test below detects. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{v}_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

Are $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{v}_3$ linearly independent?

Hint

Compare $\underset{\sim}{v}_3$ with $\underset{\sim}{v}_1 + \underset{\sim}{v}_2$.

Solution

Solution 0.1. No. $\underset{\sim}{v}_1 + \underset{\sim}{v}_2 = (1, 3, 1)^\top = \underset{\sim}{v}_3$, so

$$1 \cdot \underset{\sim}{v}_1 + 1 \cdot \underset{\sim}{v}_2 + (-1) \cdot \underset{\sim}{v}_3 = \underset{\sim}{0},$$

with numbers $c_1 = 1$, $c_2 = 1$ and $c_3 = -1$ that are not all 0. By Definition 0.1, the three vectors are linearly dependent.

Definition 0.2 (Rank)**.** The **rank** [2] of a matrix $\mathbf{A}$, written rank($\mathbf{A}$), is the largest number of columns of $\mathbf{A}$ that are linearly independent (Definition 0.1).

Example 0.2 (The rank of two 3×2 matrices)**.** The matrix $\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ has rank 2: if $c_1(1, 1, 1) + c_2(1, 2, 3) = \underset{\sim}{0}$, then subtracting the first entry from the second gives $c_2 = 0$, and then $c_1 = 0$.

The matrix $\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ has rank 1: its second column is twice its first, so the two columns are not linearly independent, but the first column on its own is.

Definition 0.3 (Full column rank)**.** An $n \times p$ matrix has **full column rank** if its rank (Definition 0.2) is p, so that all of its columns are linearly independent.

Remark (Full column rank needs at least as many rows as columns)

Remark 0.1 (Full column rank needs at least as many rows as columns). In Example 0.2, the first matrix has full column rank: its rank is 2, and it has 2 columns. The second does not, because its rank is 1.

An $n \times p$ matrix can have full column rank only when $p \leq n$, because more than n vectors of length n are never linearly independent. For example, the 2×3 matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ has 3 columns of length 2, and $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = \underset{\sim}{0}$, with coefficients that are not all zero. So its 3 columns are not linearly independent, and its rank is less than 3.

Exercise 0.2 (Finding the rank of a 3×3 matrix)**.** The rank of a data matrix counts how many of its columns carry new information. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}.$$

Find rank($\mathbf{A}$).

Hint

Add the first two columns of $\mathbf{A}$ and compare them with the third. Then check whether the first two are linearly independent.

Solution

Solution 0.2. rank($\mathbf{A}$) = 2. The third column of $\mathbf{A}$ adds up the first two: $(1,0,2)^\top + (0,1,1)^\top = (1,1,3)^\top$, so the three are not linearly independent. The first two are: if $c_1(1,0,2)^\top + c_2(0,1,1)^\top = \underset{\sim}{0}$, then the first entry gives $c_1 = 0$, and the second gives $c_2 = 0$. By Definition 0.2, the rank is 2.

Exercise 0.3 (Testing whether an all-ones column and one feature have full column rank)**.** Regression software often puts an all-ones column in the input matrix, beside the features. With one feature, measured at 2, 4 and 6, the input matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{bmatrix}.$$

Does $\mathbf{X}$ have full column rank?

Hint

By Definition 0.3, check whether the two columns of $\mathbf{X}$ are linearly independent.

Solution

Solution 0.3. Yes. Suppose $c_1(1,1,1)^\top + c_2(2,4,6)^\top = \underset{\sim}{0}$. The first two entries give

$$c_1 + 2c_2 = 0 \quad \text{and} \quad c_1 + 4c_2 = 0.$$

Subtracting the first equation from the second gives $2c_2 = 0$, so $c_2 = 0$, and then $c_1 = 0$. So the two columns of $\mathbf{X}$ are linearly independent, the rank of $\mathbf{X}$ is 2, and $\mathbf{X}$ has full

column rank.

Subspaces

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Definition 0.1 (Subspace, linear subspace)**.** A set $\mathcal{S}$ of vectors in $\mathbb{R}^p$ is a **subspace** [4] of $\mathbb{R}^p$ if it is not empty and it is closed under vector addition and scalar multiplication:

1. if $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{S}$, then $\underset{\sim}{u} + \underset{\sim}{v} \in \mathcal{S}$;
2. if $\underset{\sim}{u} \in \mathcal{S}$ and $c \in \mathbb{R}$, then $c\underset{\sim}{u} \in \mathcal{S}$.

A subspace is also called a **linear subspace**.

Example 0.1 (A line through the origin is a subspace)**.** Let $\mathcal{S} = \{c\,(1, 2) : c \in \mathbb{R}\}$, the line in $\mathbb{R}^2$ through $(0, 0)$ and $(1, 2)$.

- **Addition:** $a\,(1,2) + b\,(1,2) = (a+b)\,(1,2)$, which is in $\mathcal{S}$.
- **Scalar multiplication:** $c\,(a\,(1,2)) = (ca)\,(1,2)$, which is in $\mathcal{S}$.

So $\mathcal{S}$ is a subspace of $\mathbb{R}^2$. Two other subspaces of $\mathbb{R}^p$ are the set $\{\underset{\sim}{0}\}$ that holds only the zero vector, and $\mathbb{R}^p$ itself.

Example 0.2 (A line that misses the origin is not a subspace)**.** Let $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$, the horizontal line in $\mathbb{R}^2$ at height 1. The vectors $(0, 1)$ and $(2, 1)$ are in $\mathcal{T}$, but their sum $(2, 2)$ is not, because its second entry is 2, not 1. So $\mathcal{T}$ is not closed under addition, and it is not a subspace.

Exercise 0.1 (Testing whether vectors with a nonnegative second entry are a subspace)**.** A constraint such as "this entry is at least 0" cuts out a set of vectors, and it is worth knowing whether that set is a subspace. Let

$$\mathcal{H} = \{(x, y) \in \mathbb{R}^2 : y \geq 0\},$$

the vectors whose second entry is nonnegative.
Is $\mathcal{H}$ a subspace of $\mathbb{R}^2$?

Hint

Check condition 2 of Definition 0.1 with $c = -1$.

Solution

Solution 0.1. No. The vector $(0, 1)^\top$ is in $\mathcal{H}$, but $(-1) \cdot (0, 1)^\top = (0, -1)^\top$ is not, because its second entry is negative. So $\mathcal{H}$ fails condition 2 of Definition 0.1, and it is not a subspace. $\mathcal{H}$ does satisfy condition 1, which shows that one condition alone is not enough.

Theorem 0.1 (Every subspace contains the zero vector)**.** *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.1), then $\underset{\sim}{0} \in \mathcal{S}$.*

Proof

Proof. A subspace is not empty, so it contains some vector $\underset{\sim}{u}$. Closure under scalar multiplication with $c = 0$ puts $0 \cdot \underset{\sim}{u}$ in $\mathcal{S}$, and $0 \cdot \underset{\sim}{u} = \underset{\sim}{0}$. □

Example 0.3 (Using the zero vector to rule out a subspace)**.** The plane $\{(x, y, z) : x + y + z = 1\}$ in $\mathbb{R}^3$ does not contain $\underset{\sim}{0} = (0, 0, 0)$, because $0 + 0 + 0 = 0 \neq 1$. So, by Theorem 0.1, it is not a subspace. The line $\mathcal{T}$ in Example 0.2 fails the same test: $(0, 0)$ is not in $\mathcal{T}$, because its second entry is not 1.
Containing $\underset{\sim}{0}$ is necessary but not sufficient. The union of the two coordinate axes in $\mathbb{R}^2$, $\{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$, contains $\underset{\sim}{0}$, but $(1, 0) + (0, 1) = (1, 1)$ lies on neither axis, so the union is not closed under addition and is not a subspace.

Exercise 0.2 (Testing whether $p_1 + p_2 = 1$ defines a subspace)**.** A coin toss can land heads or tails, and the probabilities p_1 and p_2 of those two results satisfy $p_1 + p_2 = 1$. That condition defines the set

$$\mathcal{P} = \{(p_1, p_2) \in \mathbb{R}^2 : p_1 + p_2 = 1\}.$$

Is $\mathcal{P}$ a subspace of $\mathbb{R}^2$?

Hint

Use Theorem 0.1.

Solution

Solution 0.2. No. The zero vector $(0, 0)^\top$ is not in $\mathcal{P}$, because $0 + 0 = 0 \neq 1$. By Theorem 0.1, every subspace contains $\underset{\sim}{0}$, so $\mathcal{P}$ is not a subspace.

Definition 0.2 (Span)**.** The **span** [5] of vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ is the set of all their linear combinations:

$$\operatorname{span}\{\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k\} \stackrel{\text{def}}{=} \{c_1\underset{\sim}{v}_1 + \cdots + c_k\underset{\sim}{v}_k : c_1, \ldots, c_k \in \mathbb{R}\}.$$

The vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ **span** a set $\mathcal{S}$ if their span equals $\mathcal{S}$.
By convention, the span of the empty list ($k = 0$) is $\{\underset{\sim}{0}\}$, and the empty list counts as linearly independent (Definition 0.1), since it has no coefficients that could fail to be zero.

Example 0.4 (Two spans in $\mathbb{R}^3$)**.**

- $\operatorname{span}\{(1, 0, 0), (0, 1, 0)\}$ is the set of vectors $(c_1, c_2, 0)$, the plane $z = 0$ in $\mathbb{R}^3$.
- $\operatorname{span}\{(1, 2, 3), (2, 4, 6)\}$ is only the line $\{c\,(1, 2, 3) : c \in \mathbb{R}\}$, because $(2, 4, 6) = 2\,(1, 2, 3)$, so $c_1(1, 2, 3) + c_2(2, 4, 6) = (c_1 + 2c_2)\,(1, 2, 3)$. Adding a vector to a list does not always enlarge its span.

Exercise 0.3 (Testing whether a vector lies in the span of two others)**.** Whether a target can be written as a linear combination of a few given vectors is a question about a span. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{w} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}.$$

Is the vector $\underset{\sim}{w}$ in $\operatorname{span}\{\underset{\sim}{v}_1, \underset{\sim}{v}_2\}$?

Hint

Look for numbers c_1 and c_2 with $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = \underset{\sim}{w}$, one entry at a time.

Solution

Solution 0.3. Yes. Entry by entry, $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = (c_1,\ c_1 + c_2,\ c_2)^\top$. Matching $\underset{\sim}{w}$, the first entry gives $c_1 = 3$ and the third gives $c_2 = 2$. Then the second entry is $3 + 2 = 5$, which matches too, so $\underset{\sim}{w} = 3\underset{\sim}{v}_1 + 2\underset{\sim}{v}_2$ is in the span.

Theorem 0.2 (A span is a subspace)**.** *For any vectors $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k \in \mathbb{R}^p$, $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$ (Definition 0.2) is a subspace of $\mathbb{R}^p$ (Definition 0.1).*

Proof

Proof. The span is not empty: it contains $0 \cdot \underset{\sim}{v}_1 + \cdots + 0 \cdot \underset{\sim}{v}_k = \underset{\sim}{0}$.
Addition. Take two vectors in the span, $\underset{\sim}{u} = a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k$ and $\underset{\sim}{w} = b_1\underset{\sim}{v}_1 + \cdots + b_k\underset{\sim}{v}_k$. Then

$$\begin{aligned} \underset{\sim}{u} + \underset{\sim}{w} &= (a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k) + (b_1\underset{\sim}{v}_1 + \cdots + b_k\underset{\sim}{v}_k) && \text{(substitute the two combinations)} \\ &= (a_1 + b_1)\,\underset{\sim}{v}_1 + \cdots + (a_k + b_k)\,\underset{\sim}{v}_k && \text{(regroup, and distribute each } \underset{\sim}{v}_i\text{)} \end{aligned}$$

which is a linear combination of $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$, so it is in the span.
Scalar multiplication. For $c \in \mathbb{R}$,

$$\begin{aligned} c\,\underset{\sim}{u} &= c\,(a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k) && \text{(substitute the combination)} \\ &= (ca_1)\,\underset{\sim}{v}_1 + \cdots + (ca_k)\,\underset{\sim}{v}_k && \text{(distribute } c \text{ over the sum)} \end{aligned}$$

which is again in the span. □

Example 0.5 (The line in Example 0.1 is a span)**.** The line $\mathcal{S} = \{c\,(1,2) : c \in \mathbb{R}\}$ in Example 0.1 is span$\{(1,2)\}$, so Theorem 0.2 gives a second proof that it is a subspace. Likewise, the plane $z = 0$ in Example 0.4 is a subspace of $\mathbb{R}^3$, because it is span$\{(1,0,0),(0,1,0)\}$.

Exercise 0.4 (Recognizing a set of vectors as a span)**.** Describing a set by free parameters often shows that it is a span. Let

$$\mathcal{S} = \{(a+b,\ a-b,\ 2a) : a, b \in \mathbb{R}\}.$$

Is $\mathcal{S}$ a subspace of $\mathbb{R}^3$?

Hint

Write $(a+b,\ a-b,\ 2a)$ as a times one vector plus b times another, then use Theorem 0.2.

Solution

Solution 0.4. Yes. For every $a, b \in \mathbb{R}$,

$$\begin{bmatrix} a+b \\ a-b \\ 2a \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

so $\mathcal{S} = \text{span}\left\{(1,1,2)^\top, (1,-1,0)^\top\right\}$. By Theorem 0.2, $\mathcal{S}$ is a subspace of $\mathbb{R}^3$.

Theorem 0.3 (A matrix-vector product combines the columns)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix with columns* $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ *and* $\underset{\sim}{x} \in \mathbb{R}^n$*, then*

$$\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n.$$

Proof

Proof. Compare entry i of the two sides, for each $i = 1, \ldots, m$. Entry i of $\underset{\sim}{a}_j$ is a_{ij}, so

$$\begin{aligned} (\mathbf{A}\underset{\sim}{x})_i &= a_{i1}x_1 + \cdots + a_{in}x_n && \text{(Definition 16 in Matrices)} \\ &= x_1a_{i1} + \cdots + x_na_{in} && \text{(multiplication of numbers is commutative)} \\ &= (x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n)_i && \text{(entry } i \text{ of each } x_j\underset{\sim}{a}_j \text{ is } x_ja_{ij}) \end{aligned}$$

□

Example 0.6 (A product as a combination of columns)**.**

$$\begin{aligned}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}1\\ -1\end{bmatrix} &= 1\cdot\begin{bmatrix}1\\ 3\end{bmatrix} + (-1)\cdot\begin{bmatrix}2\\ 4\end{bmatrix} && \text{(Theorem 0.3)}\\ &= \begin{bmatrix}-1\\ -1\end{bmatrix}, && \text{(arithmetic)}\end{aligned}$$

the same answer as computing each entry as a row-times-vector dot product (Remark 4 in Matrices).

Exercise 0.5 (Computing a matrix-vector product as a combination of columns)**.** A linear model's predictions for every observation at once are a matrix-vector product $\mathbf{A}\underset{\sim}{x}$, and Theorem 0.3 writes it as a combination of the columns of $\mathbf{A}$. Let

$$\mathbf{A} = \begin{bmatrix}2 & 1\\ 0 & 3\\ 1 & -1\end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix}3\\ 2\end{bmatrix}.$$

Find $\mathbf{A}\underset{\sim}{x}$ with Theorem 0.3.

Hint

Use Theorem 0.3 with $x_1 = 3$ and $x_2 = 2$.

Solution

Solution 0.5. By Theorem 0.3,

$$\mathbf{A}\underset{\sim}{x} = 3\begin{bmatrix}2\\ 0\\ 1\end{bmatrix} + 2\begin{bmatrix}1\\ 3\\ -1\end{bmatrix} = \begin{bmatrix}6+2\\ 0+6\\ 3-2\end{bmatrix} = \begin{bmatrix}8\\ 6\\ 1\end{bmatrix}.$$

Theorem 0.4 (A matrix maps a combination to the same combination of images)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix,* $\underset{\sim}{z}_1, \dots, \underset{\sim}{z}_k \in \mathbb{R}^n$ *and* $c_1, \dots, c_k \in \mathbb{R}$*, then*

$$\mathbf{A}\left(c_1\underset{\sim}{z}_1 + \cdots + c_k\underset{\sim}{z}_k\right) = c_1\,\mathbf{A}\underset{\sim}{z}_1 + \cdots + c_k\,\mathbf{A}\underset{\sim}{z}_k.$$

Proof

Proof. The function $\underset{\sim}{x} \mapsto \mathbf{A}\underset{\sim}{x}$ is a linear map by Theorem 12 in Matrices, and a linear map preserves linear combinations by Theorem 13 in Matrices. Applying that theorem

with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ and $\underset{\sim}{v}_i = \underset{\sim}{z}_i$ gives the identity. □

Example 0.7 (Applying a matrix to a combination)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\underset{\sim}{z}_1 = (1, 0)$ and $\underset{\sim}{z}_2 = (0, 1)$. Then $\mathbf{A}\underset{\sim}{z}_1 = (1, 3)$ and $\mathbf{A}\underset{\sim}{z}_2 = (2, 4)$, and for the combination $2\underset{\sim}{z}_1 - \underset{\sim}{z}_2 = (2, -1)$:

$$\begin{aligned} \mathbf{A}\,(2, -1) &= (1 \cdot 2 + 2 \cdot (-1),\ 3 \cdot 2 + 4 \cdot (-1)) && \text{(Definition 16 in Matrices)} \\ &= (0, 2) && \text{(arithmetic)} \\ &= 2\,(1, 3) - (2, 4), && \text{(arithmetic)} \end{aligned}$$

which is $2\,\mathbf{A}\underset{\sim}{z}_1 - \mathbf{A}\underset{\sim}{z}_2$, as Theorem 0.4 says.

Exercise 0.6 (Applying a matrix to a combination without knowing the matrix)**.** Theorem 0.4 lets a matrix-vector product of a combination be found from the matrix-vector product of each part, even when the matrix itself is not known. Suppose that a matrix $\mathbf{A}$ satisfies

$$\mathbf{A}\underset{\sim}{z}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{A}\underset{\sim}{z}_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Find $\mathbf{A}\,(3\underset{\sim}{z}_1 + 2\underset{\sim}{z}_2)$.

Hint

Use Theorem 0.4 with $c_1 = 3$ and $c_2 = 2$.

Solution

Solution 0.6. By Theorem 0.4,

$$\mathbf{A}\,(3\underset{\sim}{z}_1 + 2\underset{\sim}{z}_2) = 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2\begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Theorem 0.5 (More than p vectors in $\mathbb{R}^p$ are linearly dependent)**.** *If $k > p$, then any k vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ are not linearly independent (Definition 0.1): some numbers $c_1, \ldots, c_k$, not all zero, satisfy $c_1\underset{\sim}{v}_1 + \cdots + c_k\underset{\sim}{v}_k = \underset{\sim}{0}_p$.*

Proof

Proof. Induction on p.
Base case, $p = 1$**.** Each $\underset{\sim}{v}_i$ is a single number v_i, and $k \geq 2$. If $v_1 = 0$, take $c_1 = 1$ and every other $c_i = 0$. Otherwise take $c_1 = v_2$, $c_2 = -v_1$ and every other $c_i = 0$; then $c_2 \neq 0$, and $v_2 v_1 - v_1 v_2 = 0$.
Inductive step. Let $p \geq 2$, assume the theorem holds for vectors in $\mathbb{R}^{p-1}$, and take $k > p$ vectors in $\mathbb{R}^p$. Write $v_{i,p}$ for the last entry of $\underset{\sim}{v}_i$.
Case 1: every $v_{i,p} = 0$. Deleting the last entry gives $k > p-1$ vectors in $\mathbb{R}^{p-1}$, which by the inductive hypothesis have a combination with coefficients not all zero that equals $\underset{\sim}{0}_{p-1}$. The same coefficients combine the original vectors to $\underset{\sim}{0}_p$, since every last entry is 0.
Case 2: some $v_{i,p} \neq 0$. Renumber the vectors so that $v_{k,p} \neq 0$, and for $i = 1, \ldots, k-1$ let

$$\underset{\sim}{w}_i \stackrel{\text{def}}{=} \underset{\sim}{v}_i - \frac{v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k.$$

The last entry of $\underset{\sim}{w}_i$ is $v_{i,p} - \frac{v_{i,p}}{v_{k,p}} v_{k,p} = 0$. Deleting that zero last entry gives $k-1 > p-1$ vectors in $\mathbb{R}^{p-1}$, so by the inductive hypothesis there are numbers $c_1, \ldots, c_{k-1}$, not all zero, whose combination of the shortened vectors is $\underset{\sim}{0}_{p-1}$. Putting the zero last entries back gives $c_1 \underset{\sim}{w}_1 + \cdots + c_{k-1} \underset{\sim}{w}_{k-1} = \underset{\sim}{0}_p$. Then

$$\begin{aligned}
\underset{\sim}{0}_p &= c_1 \underset{\sim}{w}_1 + \cdots + c_{k-1} \underset{\sim}{w}_{k-1} && \text{(choice of } c_1, \ldots, c_{k-1}\text{)} \\
&= \sum_{i=1}^{k-1} c_i \left(\underset{\sim}{v}_i - \frac{v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k \right) && \text{(definition of } \underset{\sim}{w}_i\text{)} \\
&= \sum_{i=1}^{k-1} c_i \underset{\sim}{v}_i - \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k && \text{(distribute each } c_i\text{, and distribute the sum over the difference} \\
&= c_1 \underset{\sim}{v}_1 + \cdots + c_{k-1} \underset{\sim}{v}_{k-1} - \left(\sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \right) \underset{\sim}{v}_k, && \text{(factor } \underset{\sim}{v}_k \text{ out of the second sum)}
\end{aligned}$$

a combination of $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ equal to $\underset{\sim}{0}_p$ whose first $k-1$ coefficients are not all zero. □

Example 0.8 (Three vectors in $\mathbb{R}^2$)**.** Theorem 0.5 says the three vectors $(1,0)$, $(0,1)$ and $(1,1)$ in $\mathbb{R}^2$ are linearly dependent, and indeed $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) = (0,0)$. The theorem says nothing about $k \leq p$ vectors: $(1,0)$ and $(0,1)$ are linearly independent, while $(1,2)$ and $(2,4)$ are not (Example 0.1).

Exercise 0.7 (Testing four vectors in $\mathbb{R}^3$ for linear independence)**.** A table of data with more features than observations has more columns than each column has entries, and

Theorem 0.5 settles such cases without any arithmetic. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \underset{\sim}{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \underset{\sim}{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 10 \end{bmatrix}, \quad \underset{\sim}{v}_4 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Are these four vectors in $\mathbb{R}^3$ linearly independent?

Hint

Compare the number of vectors with $p = 3$ in Theorem 0.5.

Solution

Solution 0.7. No. There are $k = 4$ vectors in $\mathbb{R}^3$, and $4 > 3$, so by Theorem 0.5 they are linearly dependent.

Definition 0.3 (Basis)**.** A **basis** [6] of a subspace $\mathcal{S}$ (Definition 0.1) is a list of vectors that are linearly independent (Definition 0.1) and span $\mathcal{S}$ (Definition 0.2).

Example 0.9 (Two bases of $\mathbb{R}^2$)**.**

- $(1, 0)$ and $(0, 1)$ form a basis of $\mathbb{R}^2$: they are linearly independent, and any $(x, y) \in \mathbb{R}^2$ equals $x\,(1, 0) + y\,(0, 1)$.
- $(1, 0)$ and $(1, 1)$ also form a basis of $\mathbb{R}^2$: Example 0.1 shows they are linearly independent, and any (x, y) equals $(x - y)\,(1, 0) + y\,(1, 1)$, because $(x - y) + y = x$ in the first entry and $0 + y = y$ in the second.

Example 0.10 (Lists that are not bases of $\mathbb{R}^2$)**.**

- $(1, 0)$ and $(2, 0)$ are not a basis of $\mathbb{R}^2$. They are not linearly independent, since $2\,(1, 0) - (2, 0) = \underset{\sim}{0}$, and they do not span $\mathbb{R}^2$: every combination $c_1(1, 0) + c_2(2, 0)$ has second entry $\widetilde{0}$, so $(0, 1)$ is not in their span.
- $(1, 0)$, $(0, 1)$ and $(1, 1)$ span $\mathbb{R}^2$, but they are not a basis: $(1, 0) + (0, 1) - (1, 1) = \underset{\sim}{0}$, so they are not linearly independent.

Exercise 0.8 (Testing whether $(1, 1)$ and $(1, -1)$ form a basis of $\mathbb{R}^2$)**.** Describing every vector in $\mathbb{R}^2$ through $(1, 1)$ and $(1, -1)$ instead of the axes works only if those two vectors form a basis. Let

$$\underset{\sim}{a}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{a}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Is $\underset{\sim}{a}_1, \underset{\sim}{a}_2$ a basis of $\mathbb{R}^2$?

Hint

Check both conditions of Definition 0.3. For spanning, look for c_1 and c_2 with $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (x, y)^\top$.

Solution

Solution 0.8. Yes. They span $\mathbb{R}^2$: for every $(x, y)^\top \in \mathbb{R}^2$,

$$\frac{x+y}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{x-y}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

They are linearly independent: if $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = \underset{\sim}{0}$, then $c_1 + c_2 = 0$ and $c_1 - c_2 = 0$, and adding the two equations gives $c_1 = 0$ and then $c_2 = 0$. By Definition 0.3, $\underset{\sim}{a}_1, \underset{\sim}{a}_2$ is a basis of $\mathbb{R}^2$.

Theorem 0.6 (Coefficients in a basis are unique)**.** *If $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ is a basis of a subspace $\mathcal{S}$ (Definition 0.3), then every $\underset{\sim}{x} \in \mathcal{S}$ is a linear combination $\underset{\sim}{x} = c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k$ for exactly one list of numbers $c_1, \ldots, c_k$.*

Proof

Proof. At least one such list exists, because the basis spans $\mathcal{S}$ (Definition 0.2). Suppose two lists both work:

$$\begin{aligned} c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k &= \underset{\sim}{x} \\ &= d_1\underset{\sim}{a}_1 + \cdots + d_k\underset{\sim}{a}_k. \end{aligned}$$

Subtracting the right side from the left side gives

$$\begin{aligned} \underset{\sim}{0} &= (c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k) - (d_1\underset{\sim}{a}_1 + \cdots + d_k\underset{\sim}{a}_k) && \text{(both combinations equal } \underset{\sim}{x}\text{)} \\ &= (c_1 - d_1)\,\underset{\sim}{a}_1 + \cdots + (c_k - d_k)\,\underset{\sim}{a}_k && \text{(regroup, and distribute each } \underset{\sim}{a}_i\text{)} \end{aligned}$$

Because $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ are linearly independent (Definition 0.1), every coefficient $c_i - d_i$ is 0, so $c_i = d_i$ for each i. □

Example 0.11 (Coefficients of $(3,5)$ in two bases)**.** In the basis $(1,0),(0,1)$ of Example 0.9, $(3,5) = 3\,(1,0) + 5\,(0,1)$, and no other coefficients work. In the basis $(1,0),(1,1)$, the formula in Example 0.9 gives

$$\begin{aligned}(3,5) &= (3-5)\,(1,0) + 5\,(1,1)\\ &= -2\,(1,0) + 5\,(1,1).\end{aligned}$$

The same vector has different coefficients in different bases, but within one basis its coefficients are unique.
The list $(1,0),(0,1),(1,1)$ in Example 0.10 spans $\mathbb{R}^2$ but is not linearly independent, and the coefficients of $(3,5)$ are not unique:

$$\begin{aligned}(3,5) &= 3\,(1,0) + 5\,(0,1) + 0\,(1,1)\\ &= 0\,(1,0) + 2\,(0,1) + 3\,(1,1).\end{aligned}$$

Exercise 0.9 (Counting the ways to write a vector in a basis)**.** The vectors $\underset{\sim}{a}_1 = (1,1)^\top$ and $\underset{\sim}{a}_2 = (1,-1)^\top$ form a basis of $\mathbb{R}^2$, and

$$\begin{bmatrix}5\\1\end{bmatrix} = 3\underset{\sim}{a}_1 + 2\underset{\sim}{a}_2.$$

How many lists of numbers c_1, c_2 satisfy $(5,1)^\top = c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$?

Hint

Use Theorem 0.6.

Solution

Solution 0.9. Exactly one, namely $c_1 = 3$ and $c_2 = 2$. By Theorem 0.6, a vector in the span of a basis has only one list of numbers, and $3, 2$ is a list that works.

Definition 0.4 (Coordinates in a basis)**.** Let $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$ be a basis of a subspace $\mathcal{S}$ (Definition 0.3), and let $\underset{\sim}{x} \in \mathcal{S}$. The **coordinates** [7] of $\underset{\sim}{x}$ in this basis are the numbers $c_1, \dots, c_k$ with $\underset{\sim}{x} = c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k$. By Theorem 0.6 there is exactly one such list of numbers.

Definition 0.5 (Coordinate vector)**.** With $\underset{\sim}{x}$ and its coordinates $c_1, \dots, c_k$ as in Definition 0.4, $(c_1, \dots, c_k) \in \mathbb{R}^k$ is the **coordinate vector** of $\underset{\sim}{x}$.

Example 0.12 (Coordinates of $(4,1)$ in two bases of $\mathbb{R}^2$)**.** In the basis $(1,0),(1,1)$ of Example 0.9, the coordinates of $(4,1)$ are 3 and 1:

$$\begin{aligned} 3\,(1,0)+1\,(1,1) &= (3+1,\ 0+1) && \text{(Definition 9 in Vectors, entry by entry)} \\ &= (4,1). && \text{(add)} \end{aligned}$$

In the basis $(1,0),(0,1)$, its coordinates are its entries, 4 and 1.

Exercise 0.10 (Finding coordinates in a basis other than the indicator vectors)**.** The coordinates of a vector depend on the basis, and they match its entries only in the basis of indicator vectors. The vectors $\underset{\sim}{a}_1 = (2,0)^\top$ and $\underset{\sim}{a}_2 = (1,1)^\top$ form a basis of $\mathbb{R}^2$. Let

$$\underset{\sim}{x} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}.$$

Find the coordinates of $\underset{\sim}{x}$ in the basis $\underset{\sim}{a}_1, \underset{\sim}{a}_2$.

Hint

Write $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$ entry by entry, and start from the second entry.

Solution

Solution 0.10. The coordinates of $\underset{\sim}{x}$ are 2 and 3. Entry by entry, $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (2c_1+c_2,\ c_2)^\top$. The second entry gives $c_2 = 3$, and then the first gives $2c_1 + 3 = 7$, so $c_1 = 2$. As a check, $2(2,0)^\top + 3(1,1)^\top = (7,3)^\top$.

Exercise 0.11 (Describing a point on a plane in $\mathbb{R}^3$ with two numbers)**.** A point on a plane through the origin in $\mathbb{R}^3$ needs only two numbers, its coordinate vector in a basis of the plane. The vectors $\underset{\sim}{a}_1 = (1,1,0)^\top$ and $\underset{\sim}{a}_2 = (0,1,1)^\top$ form a basis of a plane $\mathcal{S}$, which contains

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Find the coordinate vector of $\underset{\sim}{x}$ in the basis $\underset{\sim}{a}_1, \underset{\sim}{a}_2$.

Hint

Write $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$ entry by entry, then use Definition 0.5.

Solution

Solution 0.11. Entry by entry, $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (c_1,\ c_1 + c_2,\ c_2)^\top$. Matching $\underset{\sim}{x}$ gives $c_1 = 2$ and $c_2 = 3$, and the second entry checks: $2 + 3 = 5$. By Definition 0.5, the coordinate vector of $\underset{\sim}{x}$ is $(2, 3)^\top \in \mathbb{R}^2$.

Theorem 0.7 (All bases of a subspace have the same number of vectors)**.** *If* $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$ *and* $\underset{\sim}{b}_1, \dots, \underset{\sim}{b}_l$ *are both bases of a subspace* $\mathcal{S}$ *of* $\mathbb{R}^p$ *(Definition 0.3), then* $k = l$*.*

Proof

Proof. Let $\mathbf{A}$ be the $p \times k$ matrix with columns $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$, and $\mathbf{B}$ the $p \times l$ matrix with columns $\underset{\sim}{b}_1, \dots, \underset{\sim}{b}_l$.

Write $\mathbf{A}$ in terms of $\mathbf{B}$. Each $\underset{\sim}{a}_j$ is in $\mathcal{S}$, and the $\underset{\sim}{b}$'s span $\mathcal{S}$ (Definition 0.2), so $\underset{\sim}{a}_j = \mathbf{B}\underset{\sim}{c}_j$ for some $\underset{\sim}{c}_j \in \mathbb{R}^l$ (Definition 16 in Matrices). Let $\mathbf{C}$ be the $l \times k$ matrix with columns $\underset{\sim}{c}_1, \dots, \underset{\sim}{c}_k$. Then $\underset{p\times k}{\mathbf{A}} = \underset{p\times l}{\mathbf{B}}\ \underset{l\times k}{\mathbf{C}}$, because column j of $\mathbf{BC}$ is $\mathbf{B}\underset{\sim}{c}_j$.

The columns of $\mathbf{C}$ are linearly independent. Suppose $\mathbf{C}\underset{\sim}{x} = \underset{\sim}{0}$ for some $\underset{\sim}{x} \in \mathbb{R}^k$. Then

$$\begin{aligned}
\mathbf{A}\underset{\sim}{x} &= (\mathbf{BC})\,\underset{\sim}{x} && \text{(substitute } \mathbf{A} = \mathbf{BC}\text{)}\\
&= \mathbf{B}\,(\mathbf{C}\underset{\sim}{x}) && \text{(Theorem 7 in Matrices)}\\
&= \mathbf{B}\,\underset{\sim}{0} && \text{(substitute } \mathbf{C}\underset{\sim}{x} = \underset{\sim}{0}\text{)}\\
&= \underset{\sim}{0} && \text{(a matrix times the zero vector is } \underset{\sim}{0}\text{)}
\end{aligned}$$

$\mathbf{A}\underset{\sim}{x}$ is the combination $x_1\underset{\sim}{a}_1 + \cdots + x_k\underset{\sim}{a}_k$ (Theorem 0.3), and the $\underset{\sim}{a}$'s are linearly independent (Definition 0.1), so $\underset{\sim}{x} = \underset{\sim}{0}$. So the only combination of the columns of $\mathbf{C}$ that equals $\underset{\sim}{0}$ has all coefficients 0.

Count. $\mathbf{C}$ has k linearly independent columns of length l. More than l vectors in $\mathbb{R}^l$ are never linearly independent (Theorem 0.5), so $k \le l$. Exchanging the roles of the two bases gives $l \le k$, so $k = l$. □

Example 0.13 (Both bases of $\mathbb{R}^2$ in Example 0.9 have two vectors)**.** The bases $(1, 0), (0, 1)$ and $(1, 0), (1, 1)$ of $\mathbb{R}^2$ in Example 0.9 both have 2 vectors, as Theorem 0.7 requires. So no list of 3 vectors is a basis of $\mathbb{R}^2$, which agrees with Example 0.10: the list $(1, 0), (0, 1), (1, 1)$ spans $\mathbb{R}^2$ but is not linearly independent.

Definition 0.6 (Dimension)**.** The **dimension** of a subspace $\mathcal{S}$, written $\dim(\mathcal{S})$, is the number of vectors in any basis of $\mathcal{S}$ (Definition 0.3). Theorem 0.7 makes this number well defined. By convention, the empty list is the basis of $\{\underset{\sim}{0}\}$, so $\dim(\{\underset{\sim}{0}\}) = 0$.

Example 0.14 (The dimensions of some subspaces)**.**

- $\dim(\{\underset{\sim}{0}\}) = 0$, by the convention in Definition 0.6.
- $\dim(\mathbb{R}^2) = 2$, by Example 0.9. More generally, $\dim(\mathbb{R}^p) = p$: the indicator vectors $\underset{\sim}{e}_1, \ldots, \underset{\sim}{e}_p$ (Definition 18 in Vectors) are linearly independent, since $c_1\underset{\sim}{e}_1 + \cdots + c_p\underset{\sim}{e}_p = (c_1, \ldots, c_p)$, which is $\underset{\sim}{0}_p$ only if every $c_i = 0$; and they span $\mathbb{R}^p$, since any $\underset{\sim}{x} = x_1\underset{\sim}{e}_1 + \cdots + x_p\underset{\sim}{e}_p$.
- The line in Example 0.1 has dimension 1: the single vector $(1, 2)$ is linearly independent and spans the line.
- The plane $z = 0$ in Example 0.4 has dimension 2, with basis $(1, 0, 0), (0, 1, 0)$.

Exercise 0.12 (Testing whether two vectors can form a basis of $\mathbb{R}^3$)**.** A list of two features cannot describe every point in $\mathbb{R}^3$, and Theorem 0.7 says why in terms of bases. The indicator vectors $\underset{\sim}{e}_1$, $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$ form a basis of $\mathbb{R}^3$.
Can a list of 2 vectors be a basis of $\mathbb{R}^3$?

Hint

Use Theorem 0.7 with the basis $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$.

Solution

Solution 0.12. No. $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$ is a basis of $\mathbb{R}^3$ with 3 vectors, so by Theorem 0.7 every basis of $\mathbb{R}^3$ has 3 vectors, not 2.

Exercise 0.13 (Finding the dimension of a span that includes a multiple)**.** The dimension of a span can be smaller than the number of vectors that span it, for example when one vector is a multiple of another. Let

$$\mathcal{S} = \operatorname{span}\left\{(1, 2, 0)^\top,\ (2, 4, 0)^\top,\ (0, 0, 1)^\top\right\}.$$

Find $\dim(\mathcal{S})$.

Hint

Look for a vector in the list that is a multiple of another, then check that the rest are linearly independent.

Solution

Solution 0.13. $\dim(\mathcal{S}) = 2$. Since $(2, 4, 0)^\top = 2(1, 2, 0)^\top$, dropping $(2, 4, 0)^\top$ leaves the span unchanged, so $(1, 2, 0)^\top$ and $(0, 0, 1)^\top$ span $\mathcal{S}$. They are linearly independent: if

$c_1(1,2,0)^\top + c_2(0,0,1)^\top = \underset{\sim}{0}$, the first entry gives $c_1 = 0$ and the third gives $c_2 = 0$. So they form a basis of $\mathcal{S}$ with 2 vectors, and by Definition 0.6 the dimension is 2.

Theorem 0.8 (Adding a vector outside the span keeps a list independent)**.** *If $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$ are linearly independent (Definition 0.1) and $\underset{\sim}{w}$ is not in* $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$ *(Definition 0.2), then $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k, \underset{\sim}{w}$ are linearly independent.*

Proof

Proof. Suppose $c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k + d\,\underset{\sim}{w} = \underset{\sim}{0}$.
The coefficient d is 0. If $d \neq 0$, then

$$\begin{aligned} \underset{\sim}{w} &= \frac{1}{d}\,(d\,\underset{\sim}{w}) && (d \neq 0) \\ &= \frac{1}{d}\,(-c_1 \underset{\sim}{v}_1 - \cdots - c_k \underset{\sim}{v}_k) && \text{(solve the supposed equation for } d\,\underset{\sim}{w}) \\ &= \left(-\frac{c_1}{d}\right) \underset{\sim}{v}_1 + \cdots + \left(-\frac{c_k}{d}\right) \underset{\sim}{v}_k, && \text{(distribute } \tfrac{1}{d}) \end{aligned}$$

which puts $\underset{\sim}{w}$ in $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$, contrary to the assumption.
The other coefficients are 0. With $d = 0$, the equation reads $c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k = \underset{\sim}{0}$, so every $c_i = 0$, because $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$ are linearly independent. □

Example 0.15 (Adding $(0,0,1)$ to two independent vectors in $\mathbb{R}^3$)**.** The vectors $(1,0,0)$ and $(0,1,0)$ are linearly independent, and their span is the plane $z = 0$ (Example 0.4). The vector $(0,0,1)$ has third entry 1, so it is not in that plane, and by Theorem 0.8 the three vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ are linearly independent. Adding $(1,1,0)$ instead would not work: $(1,1,0)$ is in the span, and $(1,0,0) + (0,1,0) - (1,1,0) = \underset{\sim}{0}$.

Exercise 0.14 (Adding a third feature to an independent pair)**.** Adding a new feature keeps a list of features linearly independent when the new one is not a combination of the old ones, which is what Theorem 0.8 says. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Are $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{w}$ linearly independent?

Hint

Check whether $\underset{\sim}{w}$ is in $\operatorname{span}\{\underset{\sim}{v}_1, \underset{\sim}{v}_2\}$, then use Theorem 0.8.

Solution

Solution 0.14. Yes. $\underset{\sim}{v}_1$ and $\underset{\sim}{v}_2$ are linearly independent: $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = (c_1, c_2, 0)^\top$ is $\underset{\sim}{0}$ only for $c_1 = c_2 = 0$. Every vector in their span has third entry 0, and $\underset{\sim}{w}$ has third entry 1, so $\underset{\sim}{w}$ is not in the span. By Theorem 0.8, $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{w}$ are linearly independent.

Theorem 0.9 (An independent list extends to a basis)**.** *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.1) and $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathcal{S}$ are linearly independent (Definition 0.1), then adding vectors of $\mathcal{S}$ to the list gives a basis of $\mathcal{S}$ (Definition 0.3). Starting from the empty list shows that every subspace has a basis.*

Proof

Proof. Repeat the following step. If the current list spans $\mathcal{S}$, stop: it is linearly independent and spans $\mathcal{S}$, so it is a basis. Otherwise, some $\underset{\sim}{w} \in \mathcal{S}$ is not in the span of the current list; add $\underset{\sim}{w}$ to the list, which stays linearly independent by Theorem 0.8.
Every list produced is a linearly independent list of vectors in $\mathbb{R}^p$, so it has at most p vectors (Theorem 0.5). Each step adds one vector, so the process stops after at most $p - k$ steps, and when it stops the list is a basis of $\mathcal{S}$.
For the empty list, which is linearly independent and spans $\{\underset{\sim}{0}\}$ by the conventions in Definition 0.2, the first step asks whether $\mathcal{S} = \{\underset{\sim}{0}\}$: if so, the empty list is already a basis; if not, the process adds vectors as described. □

Example 0.16 (Extending $(1, 1, 0)$ to a basis of $\mathbb{R}^3$)**.** Start with the single vector $(1, 1, 0)$.

1. $(1, 0, 0)$ is not in $\text{span}\{(1, 1, 0)\}$, because every multiple $c\,(1, 1, 0)$ has equal first and second entries. Add it.
2. $(0, 0, 1)$ is not in $\text{span}\{(1, 1, 0), (1, 0, 0)\}$, because every combination of those two vectors has third entry 0. Add it.
3. The list $(1, 1, 0), (1, 0, 0), (0, 0, 1)$ spans $\mathbb{R}^3$: $(x, y, z) = y\,(1, 1, 0) + (x - y)\,(1, 0, 0) + z\,(0, 0, 1)$. Stop.

The result is a basis of $\mathbb{R}^3$ that contains the starting vector.

Exercise 0.15 (Extending one vector to a basis of $\mathbb{R}^3$)**.** Theorem 0.9 says that any linearly independent list can be completed to a basis. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Find two indicator vectors that extend $\underset{\sim}{v}_1$ to a basis of $\mathbb{R}^3$.

Hint

Add indicator vectors one at a time, each outside the span of the list so far, as Theorem 0.8 allows.

Solution

Solution 0.15. One answer is $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$. The list $\underset{\sim}{v}_1$ is linearly independent, since $\underset{\sim}{v}_1 \neq \underset{\sim}{0}$. $\underset{\sim}{e}_2 = (0,1,0)^\top$ is not a multiple of $\underset{\sim}{v}_1$, whose first two entries match, so by Theorem 0.8 the list $\underset{\sim}{v}_1, \underset{\sim}{e}_2$ is linearly independent. Every vector in its span has third entry 0, so $\underset{\sim}{e}_3$ is outside it, and $\underset{\sim}{v}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$ is linearly independent too. These three vectors span $\mathbb{R}^3$, because $(x,y,z)^\top = x\underset{\sim}{v}_1 + (y-x)\underset{\sim}{e}_2 + z\underset{\sim}{e}_3$, so they form a basis of $\mathbb{R}^3$. The pair $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_3$ works too.

Theorem 0.10 (An independent list in a subspace has at most dim vectors)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = d$ (Definition 0.6).*

1. *Any linearly independent list of vectors in $\mathcal{S}$ has at most d vectors.*
2. *If $\mathcal{T}$ is a subspace with $\mathcal{T} \subseteq \mathcal{S}$, then $\dim(\mathcal{T}) \leq \dim(\mathcal{S})$.*

Proof

Proof. **Part 1.** If $d = 0$, then $\mathcal{S} = \{\underset{\sim}{0}\}$, so every vector in a nonempty list from $\mathcal{S}$ is $\underset{\sim}{0}$, and $1 \cdot \underset{\sim}{0} = \underset{\sim}{0}$ shows the list is not linearly independent; only the empty list, with $0 = d$ vectors, is. Now let $d \geq 1$, and let $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_d$ be a basis of $\mathcal{S}$, and let $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_k \in \mathcal{S}$ be linearly independent. Each $\underset{\sim}{u}_j$ is in $\mathcal{S}$, so $\underset{\sim}{u}_j = c_{1j}\underset{\sim}{a}_1 + \cdots + c_{dj}\underset{\sim}{a}_d$ for some coordinate vector $\underset{\sim}{c}_j = (c_{1j}, \ldots, c_{dj}) \in \mathbb{R}^d$. Suppose $k > d$. Then $\underset{\sim}{c}_1, \ldots, \underset{\sim}{c}_k$ are $k > d$ vectors in $\mathbb{R}^d$, so some $x_1, \ldots, x_k$, not all zero, satisfy $x_1\underset{\sim}{c}_1 + \cdots + x_k\underset{\sim}{c}_k = \underset{\sim}{0}_d$ (Theorem 0.5); entry i of that equation says $\sum_{j=1}^{k} x_j c_{ij} = 0$. Then

$$
\begin{aligned}
\sum_{j=1}^{k} x_j \underset{\sim}{u}_j &= \sum_{j=1}^{k} x_j \sum_{i=1}^{d} c_{ij} \underset{\sim}{a}_i && \text{(substitute each } \underset{\sim}{u}_j\text{)} \\
&= \sum_{j=1}^{k} \sum_{i=1}^{d} x_j c_{ij}\, \underset{\sim}{a}_i && \text{(distribute each } x_j \text{ over the inner sum)} \\
&= \sum_{i=1}^{d} \sum_{j=1}^{k} x_j c_{ij}\, \underset{\sim}{a}_i && \text{(swap the order of the finite sums)} \\
&= \sum_{i=1}^{d} \left(\sum_{j=1}^{k} x_j c_{ij} \right) \underset{\sim}{a}_i && \text{(factor } \underset{\sim}{a}_i \text{ out of the inner sum)} \\
&= \sum_{i=1}^{d} 0 \cdot \underset{\sim}{a}_i && \text{(entry } i \text{ of } \textstyle\sum_j x_j \underset{\sim}{c}_j = \underset{\sim}{0}_d\text{)} \\
&= \underset{\sim}{0}_p, && \text{(a zero multiple of a vector is } \underset{\sim}{0}\text{)}
\end{aligned}
$$

a combination of $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_k$ with coefficients not all zero that equals $\underset{\sim}{0}_p$. That contradicts their linear independence, so $k \leq d$.

Part 2. $\mathcal{T}$ has a basis (Theorem 0.9) with $\dim(\mathcal{T})$ vectors. Those vectors are linearly independent and lie in $\mathcal{S}$, so by Part 1 there are at most $\dim(\mathcal{S})$ of them. □

Example 0.17 (Subspaces of a plane have dimension at most two)**.** The plane $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$ in $\mathbb{R}^3$ has dimension 2 (Example 0.14). By Theorem 0.10, no three vectors with third entry 0 are linearly independent; for instance, $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \underset{\sim}{0}$. The line $\mathcal{T} = \text{span}\{(1, 1, 0)\}$ lies in $\mathcal{S}$, and

$$
\begin{aligned}
\dim(\mathcal{T}) &= 1 \\
&\leq 2 \\
&= \dim(\mathcal{S}).
\end{aligned}
$$

Exercise 0.16 (Testing whether four independent vectors fit in a three-dimensional subspace)**.** Theorem 0.10 caps how many linearly independent vectors a subspace can hold. Let $\mathcal{S}$ be a subspace of $\mathbb{R}^5$ with dimension 3.
Can $\mathcal{S}$ contain 4 linearly independent vectors?

Hint

Use part 1 of Theorem 0.10 with $d = 3$.

Solution

Solution 0.16. No. By part 1 of Theorem 0.10, a linearly independent list of vectors in $\mathcal{S}$ has at most $\dim(\mathcal{S}) = 3$ vectors, so it cannot have 4.

Theorem 0.11 (A subspace inside another of the same dimension equals it)**.** *If $\mathcal{T}$ and $\mathcal{S}$ are subspaces of $\mathbb{R}^p$ with $\mathcal{T} \subseteq \mathcal{S}$ and $\dim(\mathcal{T}) = \dim(\mathcal{S})$ (Definition 0.6), then $\mathcal{T} = \mathcal{S}$.*

Proof

Proof. Let $d = \dim(\mathcal{S})$, and let $\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d$ be a basis of $\mathcal{T}$ (Theorem 0.9); it has d vectors because $\dim(\mathcal{T}) = d$. (When $d = 0$ it is the empty list, whose span is $\{\underset{\sim}{0}\}$ by the convention in Definition 0.2, and the argument is unchanged.) Suppose some $\underset{\sim}{s} \in \mathcal{S}$ is not in $\operatorname{span}\{\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d\} = \mathcal{T}$. Then $\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d, \underset{\sim}{s}$ are linearly independent (Theorem 0.8), and they are $d + 1$ vectors of $\mathcal{S}$, which contradicts part 1 of Theorem 0.10. So every vector of $\mathcal{S}$ is in $\mathcal{T}$, that is, $\mathcal{S} \subseteq \mathcal{T}$. Together with $\mathcal{T} \subseteq \mathcal{S}$, this inclusion gives $\mathcal{T} = \mathcal{S}$. □

Example 0.18 (Two independent vectors in a plane span it)**.** Let $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$, the plane $z = 0$, which has dimension 2 (Example 0.14), and let $\mathcal{T} = \operatorname{span}\{(1, 0, 0), (1, 1, 0)\}$. Both vectors have third entry 0, so $\mathcal{T} \subseteq \mathcal{S}$. They are linearly independent: $c_1(1, 0, 0) + c_2(1, 1, 0) = (c_1 + c_2, c_2, 0)$, which is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$. So $\dim(\mathcal{T}) = 2$, and Theorem 0.11 gives $\mathcal{T} = \mathcal{S}$ without solving for any coefficients. As a check, $(x, y, 0) = (x - y)\,(1, 0, 0) + y\,(1, 1, 0)$.

Example 0.19 (Equal dimensions without containment)**.** The lines $\operatorname{span}\{(1, 0, 0)\}$ and $\operatorname{span}\{(0, 1, 0)\}$ in $\mathbb{R}^3$ both have dimension 1, but they are not equal: $(1, 0, 0)$ is in the first and not the second. Theorem 0.11 does not apply, because neither line contains the other.

Exercise 0.17 (Showing that a span of two vectors is all of $\mathbb{R}^2$)**.** Theorem 0.11 shows that two subspaces coincide by comparing dimensions instead of checking every vector. Let

$$\mathcal{T} = \operatorname{span}\Big\{(1, 2)^\top,\ (3, 1)^\top\Big\}.$$

Is $\mathcal{T}$ all of $\mathbb{R}^2$?

Hint

Find $\dim(\mathcal{T})$ and $\dim(\mathbb{R}^2)$, then use Theorem 0.11 with $\mathcal{S} = \mathbb{R}^2$.

Solution

Solution 0.17. Yes. The two vectors are linearly independent: if $c_1(1,2)^\top + c_2(3,1)^\top = \underset{\sim}{0}$, then $c_1 + 3c_2 = 0$ and $2c_1 + c_2 = 0$. Subtracting twice the first equation from the second gives $-5c_2 = 0$, so $c_2 = 0$ and then $c_1 = 0$. So they form a basis of $\mathcal{T}$, and $\dim(\mathcal{T}) = 2$. The indicator vectors $\underset{\sim}{e}_1, \underset{\sim}{e}_2$ form a basis of $\mathbb{R}^2$, so $\dim(\mathbb{R}^2) = 2$. Since $\mathcal{T} \subseteq \mathbb{R}^2$ and the dimensions agree, Theorem 0.11 gives $\mathcal{T} = \mathbb{R}^2$.

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